

# Randomization Tests

STAT 145

# Randomization Tests

# Today's goals

- **Set up a test.** Recognize when a statistical test is needed, and state null and alternative hypotheses in terms of a parameter.
- **Model chance with a randomization distribution** built under the assumption that the null hypothesis is true.
- **Find and interpret a  $p$ -value** from that distribution — including one- versus two-tailed tests — as the strength of evidence against the null hypothesis.
- **Make and state a decision** by comparing the  $p$ -value to a significance level, then conclude in context.
- **Know what a test can and cannot tell you:** Type I and Type II errors, the significance level as the tolerable Type I error rate, and why statistical discernibility is not the same as practical importance.

## Motivation: Cereal example

# Activity

Consider the following scenario: A sample of 80 four-to six-year olds participated in a study in which they were asked to taste two cereals, both called “Oat Bits” (in random order). The cereal in each box was exactly the same, but the boxes differed in the fact that one of the boxes had pictures of a star mascot and the other did not.



Figure 1: The two cereal boxes shown to the children. The cereal inside was identical; only one box featured a cartoon star mascot.

- a. If the cartoons make no difference, because after all the cereal is the same in each box, how many of the 80 children do you expect to pick the cartoon box? What sample proportion is this?

1. If the cartoons make no difference, each child is equally likely to pick either box.
  2. We expect about half: 40 of the 80 children.
  3. As a proportion,  $\hat{p} = 40/80 = 0.50$ .
- b. Suppose that in this study, 52 of the children chose the box with cartoons. What sample proportion is this? Does this lead you to believe that there was a preference toward the cartoon box, or would you expect this sample proportion to occur by chance even if there was no preference? (We will use StatLens to explore the sampling distribution for the proportion to determine what values are likely to occur.)

# The hypothesis testing framework

## Hypothesis testing

# Null and alternative hypotheses

## The shape of a hypothesis

| The question is about...    | Null hypothesis ("no effect") | Alternative (pick one direction)           | A question of this kind                                             |
|-----------------------------|-------------------------------|--------------------------------------------|---------------------------------------------------------------------|
| one proportion $p$          | $p = p_0$                     | $p \neq p_0, p > p_0, p < p_0$             | Do more than half of children pick the cartoon box? ( $p_0 = 0.5$ ) |
| one mean $\mu$              | $\mu = \mu_0$                 | $\mu \neq \mu_0, \mu > \mu_0, \mu < \mu_0$ | Do the bags really average the 16 oz on the label? ( $\mu_0 = 16$ ) |
| a difference in proportions | $p_1 - p_2 = 0$               | $p_1 - p_2 \neq 0, > 0, < 0$               | Do two groups differ in who skips the video?                        |
| a difference in means       | $\mu_1 - \mu_2 = 0$           | $\mu_1 - \mu_2 \neq 0, > 0, < 0$           | Is average income higher in eastern than western states?            |

# Class Example 7.1: Defining statistical hypotheses

For each of the following scenarios, state the null and alternative hypotheses with the relevant parameter(s).

a. We are testing to see if the average income is different in eastern states than western states.

1. Parameters:  $\mu_E$  and  $\mu_W$ , the mean income in eastern and western states.

2.  $H_0 : \mu_E - \mu_W = 0$  (no difference in average income).

3.  $H_A : \mu_E - \mu_W \neq 0$  (two-sided – “different” gives no direction).

b. We are testing to see if the average income is greater in eastern states than western states.

1. Parameters:  $\mu_E$  and  $\mu_W$  as above.

2.  $H_0 : \mu_E - \mu_W = 0$ .

3.  $H_A : \mu_E - \mu_W > 0$  (one-sided – “greater” points east).

c. We are testing to see if a particular coin is weighted (i.e., not fair).

1. Parameter:  $p$ , the true proportion of heads for this coin.

2.  $H_0 : p = 0.5$  (the coin is fair).

3.  $H_A : p \neq 0.5$  (two-sided – “not fair” could tip either way).

d. We are testing to see if the average rent in La Crosse is less than \$850.

# Case study: Opportunity cost

# Class Example 7.2: Opportunity cost

In the *opportunity cost* study, 150 college students each decided whether to buy a \$14.99 video. The 75 students in the **control** group chose between “buy” and “not buy,” while the 75 in the **treatment** group saw a reworded reminder — “not buy, *and keep the \$14.99 for other purchases.*” A “success” is a student who chooses *not* to buy the video. Using this study, answer the following questions:

- a. Is this an observational study or an experiment? How does the type of study impact what can be inferred from the results?
  1. It is an **experiment**: the researchers randomly assigned which version of option (B) each student saw.
  2. Because the treatment was randomly assigned, the two groups are comparable, so a discernible difference can be attributed to the phrasing — we can make a **causal** claim, not just an association.
- b. State the null and alternative hypotheses for this study.

1. Parameters:  $p_T$  and  $p_C$ , the proportion who do **not** buy the video in the treatment and control groups.
2.  $H_0 : p_T - p_C = 0$  (the reminder phrasing has no effect).
3.  $H_A : p_T - p_C > 0$  (the reminder increases the proportion who choose not to buy).

|           | Buy video | Not buy video | Total |
|-----------|-----------|---------------|-------|
| Control   | 56        | 19            | 75    |
| Treatment | 41        | 34            | 75    |
| Total     | 97        | 53            | 150   |

# Class Example 7.3: Opportunity cost, revisited

Compute the difference in the sample proportions of students who choose not to buy the video.

1. Control:  $\hat{p}_C = 19/75 = 0.253$ .
2. Treatment:  $\hat{p}_T = 34/75 = 0.453$ .
3. Difference:  $\hat{p}_T - \hat{p}_C = 0.453 - 0.253 = 0.20$  (treatment group is 20 percentage points more likely to not buy).

# Building a randomization distribution

## The card-shuffling metaphor

# Activity

Think of the data as 150 cards:

- 97 red cards (buy video)
- 53 white cards (not buy video)

In the actual study, 75 cards were dealt to the 'control' pile and 75 to the 'treatment' pile. Under the null hypothesis, the color of each card (whether to buy the video) has nothing to do with which pile it lands in.

To simulate what the null hypothesis look like, we:

1. *Thoroughly shuffle* all 150 cards together.
2. *Deal* 75 into a 'control' pile and 75 into a 'treatment' pile
3. *Count* the number of white (not buy video) in each pile and compute the difference in proportions.

This gives us one simulated difference *under the null hypothesis*, a difference that arose purely from the random shuffle, not from the phrasing.

# Many simulations

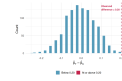


Figure 2: A histogram of 1,000 simulated differences produced under the null hypothesis. Differences at least as large as the observed 0.20 are highlighted in red.

# P-values

## Randomization Distributions and p-values

# Class Example 7.4: Estimating a $p$ -value from a randomization distribution

The figure below shows a randomization distribution for testing  $H_0 : p = 0.3$  vs  $H_A : p < 0.3$ . In each case, use the distribution to decide which value is closer to the  $p$ -value for the observed sample proportion. There are 1000 randomizations given in the figure below.

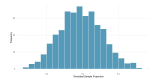


Figure 3: Randomization distribution with 1000 randomizations

a. Why is the distribution centered at 0.3?

1. The randomization distribution is built *assuming  $H_0$  is true* — that the true proportion is 0.3.
  2. Each simulated sample proportion is generated with  $p = 0.3$ , so they pile up around 0.3; the spread shows the sampling variability we get by chance alone.
- b. The  $p$ -value for  $\hat{p} = 0.15$  is closest to: 0.04 or 0.40?
1.  $H_A : p < 0.3$ , so the  $p$ -value is the *lower*-tail area — the fraction of simulations at or below 0.15.
  2. 0.15 is far into the left tail, where very few simulations land, so the  $p$ -value is small: closest to **0.04**.
- c. Which proportion,  $\hat{p} = 0.20$  or  $\hat{p} = 0.10$ , provides the strongest evidence to suggest that  $H_0$  is not true? Explain.

# Class Example 7.5: Estimating a $p$ -value from a randomization distribution...

## continued

The figure below shows a randomization distribution for testing  $H_0 : \mu_1 - \mu_2 = 0$  vs  $H_A : \mu_1 - \mu_2 \neq 0$ . The statistic used for each sample is  $D = \bar{x}_1 - \bar{x}_2$ . In each case, use the distribution to decide which value is closer to the  $p$ -value for the observed sample difference in averages. There are 1000 randomizations given in the figure below.

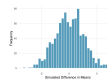


Figure 4: Randomization distribution with 1000 randomizations

a. The  $p$ -value for  $D = \bar{x}_1 - \bar{x}_2 = -2.9$  is closest to: 0.01 or 0.25?

1.  $H_A : \mu_1 - \mu_2 \neq 0$  (two-sided), so count simulations at least as far from 0 as 2.9 in either tail.
  2.  $-2.9$  is only modestly out from the center, so a fair share of simulations are that extreme: closest to **0.25**.
- b. The  $p$ -value for  $D = \bar{x}_1 - \bar{x}_2 = 1.2$  is closest to: 0.30 or 0.60?
1. 1.2 is very close to the center of the distribution, so a large majority of simulations are at least that far from 0.
  2. The two-sided  $p$ -value is large: closest to **0.60**.
- c. Of the two values of  $D$  above, which provides strongest evidence to suggest that  $H_0$  is not true? Explain.
1.  $D = -2.9$  has the smaller  $p$ -value ( $\approx 0.25$  vs.  $\approx 0.60$ ), so it sits farther in the tail.
  2. So  $D = -2.9$  gives the stronger evidence against  $H_0$  — though with  $p \approx 0.25$  neither result is discernible at  $\alpha = 0.05$ .

# Class Example 7.6: Migraine Acupuncture Trial

Can acupuncture relieve migraine pain? Allais et al. (2011) conducted a randomized controlled study where 89 individuals who suffered from migraines were randomly assigned to receive either acupuncture (treatment,  $n = 43$ ) or placebo acupuncture (control,  $n = 46$ ). The placebo acupuncture involved placing needles at non-acupuncture-point locations. Researchers recorded whether each patient was pain-free 24 hours after treatment. The full data are in the data set *Migraine Acupuncture Trial* on StatLens.

a. What are the null and alternative hypotheses for this study?

1. Parameters:  $p_T$  and  $p_C$ , the proportion pain-free after acupuncture (treatment) and sham (control).
2.  $H_0 : p_T - p_C = 0$  (acupuncture is no better than the placebo).
3.  $H_A : p_T - p_C > 0$  (acupuncture increases the proportion who are pain-free).

b. What was the sample statistic for these data?

1. Treatment:  $\hat{p}_T = 10/43 = 0.233$ . Control:  $\hat{p}_C = 2/46 = 0.043$ .
2. Statistic:  $\hat{p}_T - \hat{p}_C = 0.233 - 0.043 = 0.19$  (about 19 percentage points more pain-free in the acupuncture group).

c. What is the probability that the sample results (or those more extreme) occurred by random chance, assuming that the acupuncture had no effect?

# Class Example 7.7: Calorie Counts

Is the average amount of kilocalories from a major U.S. fast food chain menu item more than 525 kilocalories? Calorie counts for 515 menu items from eight major fast food chains were collected. The full data are in the data set *Fast Food Calories* on StatLens.

a. State the null and alternative hypotheses

1. Parameter:  $\mu$ , the mean calorie count for fast food menu items.

2.  $H_0 : \mu = 525$ .

3.  $H_A : \mu > 525$  (one-sided – “more than 525”).

b. What is the sample statistic?

1. The sample mean:  $\bar{x} \approx 531$  kilocalories (from the 515 menu items).

c. To create the randomization distribution, what do we have to assume?

1. We assume  $H_0$  is true — that the mean really is 525.
2. In StatLens we impose this by *shifting* the data so its mean sits at 525, then resampling to see how much the sample mean bounces around when  $\mu = 525$ . The distribution is therefore centered at 525.

d. What is the  $p$ -value for this study?

1. The  $p$ -value is the fraction of that null distribution at or above the observed  $\bar{x} \approx 531$ .
2. 531 sits well inside the bulk of the distribution, so the  $p$ -value is large: about **0.32**. The data do **not** provide discernible evidence that the mean exceeds 525 kcal.

# Class Example 7.8: Cereal study revisited

Recall the motivating cereal study. We were trying to determine if children had a preference for the cereal in the box with the cartoon. In the study 52 out of 80 children chose the box with cartoons. Use StatLens to find the  $p$ -value for this scenario.

1.  $H_0 : p = 0.5$  vs.  $H_A : p > 0.5$ ; observed  $\hat{p} = 52/80 = 0.65$ .
2. In StatLens, simulate many samples of 80 children with  $p = 0.5$  and find the fraction with  $\hat{p} \geq 0.65$  (52 or more choosing cartoons).
3. That upper tail is tiny: the  $p$ -value is about **0.005**. Strong evidence that children do prefer the cartoon box.

# Statistical discernibility

Decision framework

What “statistically discernible” does and does not mean

How small is small enough?

# Class Example 7.9: Red wine and weight loss

Resveratrol, an ingredient in red wine and grapes, has been shown to promote weight loss in rodents. A recent study (*Science Daily*, 2010) investigates whether the same phenomenon holds true in primates. A sample of six lemurs had their resting metabolic rate, body mass gain, food intake, and locomotor activity measured for one week prior to resveratrol supplementation and then the four indicates were measured again after treatment. For each of the following, use the  $p$ -value to make the appropriate conclusion (reject or do not reject) for the hypothesis tests. Use  $\alpha = 0.05$  for each of your decisions.

- a. In a test to see if mean resting metabolic rate is higher after treatment ( $p$ -value of 0.013).

1.  $p = 0.013 < 0.05 = \alpha$ , so **reject  $H_0$** .

b. In a test to see if mean body mass gain is lower after treatment ( $p$ -value of 0.007).

1.  $p = 0.007 < 0.05 = \alpha$ , so **reject  $H_0$** .

c. In a test to see if mean locomotor activity is affected by treatment ( $p$ -value of 0.980).

1.  $p = 0.980 \geq 0.05 = \alpha$ , so **do not reject  $H_0$** .

## Interpreting your results

# Class Example 7.10: Red wine and weight loss revisited

For each of the following, provide an interpretation of the conclusion in the context of the original study.

- a. In a test to see if mean resting metabolic rate is higher after treatment ( $p$ -value of 0.013).
  1. At  $\alpha = 0.05$ , we **do** have discernible evidence that the mean resting metabolic rate is higher after resveratrol treatment.
- b. In a test to see if mean body mass gain is lower after treatment ( $p$ -value of 0.007).
  1. At  $\alpha = 0.05$ , we **do** have discernible evidence that the mean body mass gain is lower after treatment.
- c. In a test to see if mean locomotor activity is affected by treatment ( $p$ -value of 0.980).

1. At  $\alpha = 0.05$ , we **do not** have discernible evidence that mean locomotor activity is affected by treatment.

# Class Example 7.11: Migraine Acupuncture Trial revisited

Can acupuncture relieve migraine pain? Using your decision from the Acupuncture Trial example, interpret your results in the context of the problem.

1. The  $p$ -value was about  $0.01 < 0.05$ , so we reject  $H_0$ .
2. At  $\alpha = 0.05$ , we have discernible evidence that acupuncture increases the proportion of migraine patients who are pain-free 24 hours after treatment.
3. Because patients were randomly assigned, this supports a *causal* conclusion: the acupuncture, not chance, explains the higher pain-free rate.

# Class Example 7.12: Calorie count revisited

Is the average amount of kilocalories from a major U.S. fast food chain menu item more than 525 kilocalories? Using your decision from the Calorie Count example, interpret your results in the context of the problem.

1. The  $p$ -value was about  $0.32 \geq 0.05$ , so we do not reject  $H_0$ .
2. At  $\alpha = 0.05$ , we do **not** have discernible evidence that the mean calorie count of fast food menu items exceeds 525 kilocalories.
3. Note this does not prove the mean equals 525 — only that the data are consistent with it.

# Class Example 7.13: Cereal study revisited

Recall the motivating cereal study. Interpret the results of the study in the context of the problem.

1. The  $p$ -value was about  $0.005 < 0.05$ , so we reject  $H_0$ .
2. At  $\alpha = 0.05$ , we have discernible evidence that more than half of children prefer the cereal box with cartoons.

# Type I and Type II errors

Four possible outcomes of a hypothesis test.

|                      | $H_0$ is true    | $H_A$ is true    |
|----------------------|------------------|------------------|
| Reject $H_0$         | Type I error     | Correct decision |
| Fail to reject $H_0$ | Correct decision | Type II error    |

# Class Example 7.14: Red wine and weight loss revisited

For each of the following, identify which kind of error could have occurred (Type I or Type II) using  $\alpha = 0.05$ .

a. In a test to see if mean resting metabolic rate is higher after treatment ( $p$ -value of 0.013).

1.  $p = 0.013 < 0.05$ , so we reject  $H_0$ . When we reject, the only possible error is a **Type I error** (rejecting a true  $H_0$ ).

b. In a test to see if mean body mass gain is lower after treatment ( $p$ -value of 0.007).

1.  $p = 0.007 < 0.05$ , so we reject  $H_0 \rightarrow$  a possible **Type I error**.

c. In a test to see if mean locomotor activity is affected by treatment ( $p$ -value of 0.980).

1.  $p = 0.980 \geq 0.05$ , so we do not reject  $H_0$ . When we fail to reject, the only possible error is a **Type II error** (failing to reject a false  $H_0$ ).

## What can cause these errors?

## Balancing out the errors

# Class Example 4.15: Errors in a drug study

A pharmaceutical company is researching a new anti-anxiety drug. Marketing a new drug is costly, and the company only makes a profit when the drug is effective for over 80% of patients. A random sample of 60 participants are placed on the drug and the number of participants that respond favorably to the drug will be recorded.

a. Give the null and alternative hypothesis that you would use for determining if the drug should be marketed.

1. Parameter:  $p$ , the proportion of patients who respond favorably.

2.  $H_0 : p = 0.80$  (the drug works for exactly the break-even 80%).

3.  $H_A : p > 0.80$  (worth marketing — effective for more than 80%). The burden of proof is on showing the drug clears 80%.

b. Describe in words what a Type I error would mean in the context of this study.

1. A Type I error = rejecting a true  $H_0$ : concluding the drug helps *more* than 80% of patients (and marketing it) when in truth it does not.
  2. Consequence: the company spends money marketing a drug that will not be profitable.
- c. Describe in words what a Type II error would mean in the context of this study.
1. A Type II error = failing to reject a false  $H_0$ : concluding there is not enough evidence to market the drug when it truly does help more than 80%.
  2. Consequence: the company passes on a drug that would have been effective and profitable.
- d. Suppose the significance level is set at  $\alpha = .05$ , but the researchers find that this produces a Type II error rate of 0.4 and this is too high for their liking. List two ways they could reduce the Type II error rate.

# The randomization test procedure: a summary

# Summary